Loops

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There are two types of loops:
1. Entry Control Loops
   i. while() loop
   ii. for() loop
2. Exit Control Loop
   i. do{…}while() loop

while loop

Syntax:

```
while(<condition>){
   //code to repeat
}
```

for loop

Syntax:

```
for(<initializations>;<condition;<update>){
   //code to repeat
}
```

do…while() loop

Syntax:

```
do{
  //code to repeat
}while(<condition>);
```

1. WAP to input two numbers and find out quotient and remainder without using arithmetic operators(/, %).

Input: Enter first number: 45
Enter second number: 7
Output:
Quotient: 6
Remainder: 3

Concept of divison is continous subtraction, till the result is less than or equal to the divisor

```
45
- 7
38
- 7
31
- 7
24
- 7
17
- 7
10
- 7
3
```

6 ← quotient.

remainder

By default the condition is always true.

for(; ;){ ← This is the example of infinite loop.

}

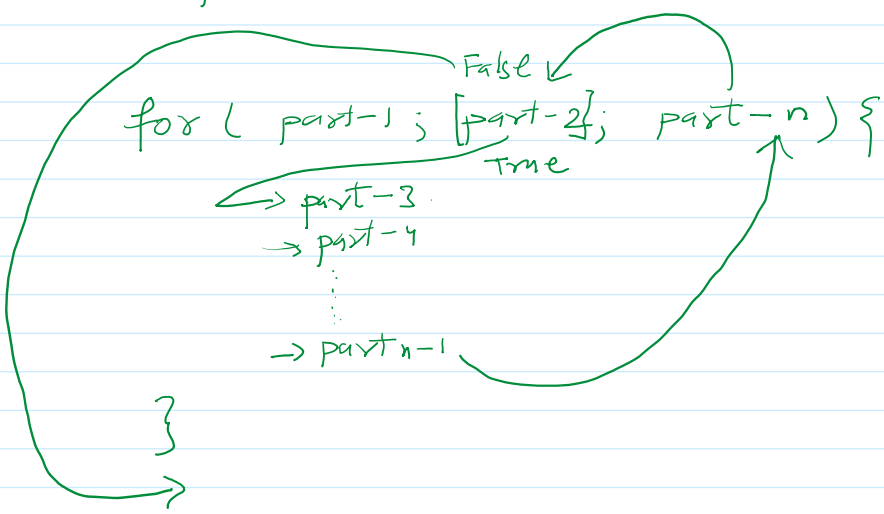
```
public class InfiniteForLoop
{
    public static void main(String[] args){
        System.out.print("\f");
        for(;;){
            System.out.print("\nHello!, I am Infinite loop");
        }
    }
}
```

optional

for (Any valid java statement ; <condition> ; Any valid java statement)
[Except, branching & looping statement] [Except, branching & looping statement]
{

// code to execute.
}

This for loop is an advance form of while loop.



2. WAP to input a number and find number of digits and sum of digits.
   Example:
   Input: Enter any number: 3456
   Output: Number of digits: 4
   Output: Sum of digits: 18